FPMANR: Dropout-Resilient and Attack-Resistant Federated Learning with Ring-Based Pairwise Masking and Layer-Wise Adaptive Noise under Non-IID Data

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ABSTRACT

Federated learning (FL) faces challenges in training stability, privacy, and secure communication, especially under non-IID data across distributed clients. Existing methods typically address optimization, privacy, or secure aggregation separately, but not all three together. This paper proposes FPMANR, an integrated FL framework that combines FedProx-based local training, layer-wise adaptive Gaussian noise, two-neighbor pairwise masking in a ring topology, and a client dropout recovery mechanism. In each communication round, 30 of 90 training clients with Dirichlet-based non-IID data train a shared global model using FedProx. Before transmission, each client adds adaptive layer-wise Gaussian noise, where larger updates receive stronger perturbation, followed by pairwise masking with two neighbors. The recovery mechanism handles client dropouts without local retraining. Experiments on EMNIST Digits, EMNIST Letters, and Fashion-MNIST compare four settings: FedProx (FP), FedProx with masking (FPM), and FPMANR with adaptive noise factors of 0.1 and 0.2 (FPMANR 0.1 and FPMANR 0.2). FPMANR 0.1 achieves test accuracies of 98.10% on EMNIST Digits, 85.39% on EMNIST Letters, and 74.51% on Fashion-MNIST. Quantitative evaluation with MSE, PSNR, SSIM, LPIPS, and reconstruction success rate shows substantially stronger resistance to DLG, iDLG, and GradInv attacks, with the success rate falling from up to 40% (FP) to 0% in all nine dataset–attack combinations. Membership inference attacks remain close to random guessing, with AUC values around 0.5 and attack accuracy near 50%. Overall, FPMANR provides a practical and attack-resistant FL framework with stable training, improved privacy, and dropout resilience under heterogeneous data. Our code is publicly available at our GitHub repository: https://github.com/Rashed200613/FPMANR.

1. Introduction

In traditional centralized machine learning (ML), data from multiple sources are transferred to a central server, where a global model is trained on the aggregated dataset (Majeed, Kaushik, Bardhan, Tadi, Min, Kumaraguru and Muni, 2022). Centralization raises regulatory, ethical, legal, and technical challenges, since anonymized electronic health records that appear GDPR- or PHI-compliant may still allow patient re-identification and information leakage (Rieke, Hancox, Li, Milletari, Roth, Albarqouni, Bakas, Galtier, Landman, Maier-Hein et al., 2020). It also causes significant communication overhead and longer training time because of continuous data and parameter transmission between clients and the server (Thangavel, Srinivasan, Naga and Narukulla, 2023). Federated learning (FL) reduces the communication cost and network congestion caused by frequent data transfers (McMahan, Moore, Ramage, Hampson and y Arcas, 2017). It enables collaborative training across many participants without sharing raw data, improving generalization while preserving privacy (Mammen, 2021).

However, FL still faces challenges under non-IID data distributions (Xiong, Cai, Takabi and Li, 2021) and remains vulnerable to gradient reconstruction attacks (Guo, Wang, Zeng, Zhu, Jiang, Wang, Zhou, Wang, Xiong and Qu, 2025) and membership inference attacks (MIAs) (Bai, Hu, Ye, Li, Wang and Xu, 2024). FedProx mitigates non-IID effects by adding a proximal term that stabilizes local training and improves convergence (Cao, 2025). Differential privacy (DP) protects user privacy by adding noise to model parameters before they are shared (Nilsson, Smith, Ulm, Gustavsson and Jirstrand, 2018), but the injected noise can reduce model performance (Zhang, Xie, Bai, Yu, Li and Gao, 2021). Adaptive Gaussian noise has been proposed to reduce the degradation caused by fixed-noise DP (Jiao, Cai, Wang, Cheng and Gao, 2023). Secure aggregation hides individual client updates from the server with cryptographic masking, but it does not fully prevent membership inference

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or data reconstruction attacks in practice (Bonawitz, Ivanov, Kreuter, Marcedone, McMahan, Patel, Ramage, Segal and Seth, 2017). Existing approaches address different aspects of FL, such as non-IID data handling, privacy preservation, communication efficiency, and robustness against inference and reconstruction attacks, but no single method resolves all of them together. This leaves a gap for FL frameworks that combine stable training under heterogeneous data, adaptive privacy for sensitive layers, and secure transmission of updates at acceptable accuracy.

Motivated by this gap, our objectives are: (1) to keep training stable under highly non-IID client data; (2) to protect individual client updates during transmission with low communication overhead; (3) to reduce information leakage from the updates with minimal loss of accuracy; (4) to keep aggregation correct when clients drop out; and (5) to evaluate the resulting privacy–utility trade-off against gradient reconstruction and membership inference attacks. To meet them, FPMAN was initially proposed in Esha, Tasnim and Das (2026) as an integrated FL framework that combines FedProx, pairwise masking, and layer-wise adaptive Gaussian noise. This paper, referred to as FPMANR, further formalizes and evaluates a client dropout recovery procedure inherent to the masking design, and extends the evaluation with broader experiments across multiple datasets and additional privacy attack analysis.

The key contributions of this study are as follows:

- We propose FPMANR, an integrated FL framework in which each client adds layer-wise adaptive Gaussian noise to its FedProx update before pairwise masking, so that masking hides individual updates from the server and the noise limits the information that remains in them (Section 3, Fig. 1).
- We introduce a two-neighbor pairwise masking scheme in a ring topology that cancels exactly in aggregation while each client communicates with only two peers, keeping per-client communication at the level of conventional FL (Section 3.4, Table 9).
- We formalize and evaluate, for the first time in this work, a dropout recovery procedure that re-pairs the nearest active neighbors from locally stored noisy updates, ensuring correct aggregation under client failures without local retraining (Section 3.5, Table 2, Fig. 5).
- We evaluate FPMANR on EMNIST Digits, EMNIST Letters, and Fashion-MNIST against DLG, iDLG, and GradInv reconstruction attacks and three membership inference attacks. FPMANR 0.1 reduces the reconstruction success rate to 0% in all nine dataset–attack combinations, with test-accuracy changes of +0.18, −5.14, and −6.06 percentage points relative to FP, and keeps MIAs near random guessing (Section 5, Tables 5–7).

The remainder of the paper is organized as follows. Section 2 reviews related work, Section 3 presents the proposed methodology and threat model, Section 4 describes the experimental setup, Section 5 reports the results, Section 6 discusses them, and Section 7 concludes the paper.

2. Related Works

This section reviews studies on non-IID optimization, differential-privacy-based FL, and secure aggregation. Table 1 summarizes the reviewed studies, their key results, and their limitations.

2.1. Non-IID Optimization: FedAvg and FedProx

The foundation of FL was established with FedAvg (McMahan et al., 2017), which enables decentralized training through local SGD and global model averaging. FedAvg reaches similar target accuracy under IID and non-IID settings (for example, 99% on MNIST and 54% on Shakespeare), but needs far more communication rounds under non-IID data (≈18–65 versus 173–600 on the MNIST CNN), showing degraded convergence due to client drift and data heterogeneity. FedProx (Li, Sahu, Zaheer, Sanjabi, Talwalkar and Smith, 2020) extends FedAvg with a proximal term in the local objective and allows partial local updates. It was evaluated on Synthetic(α,β), MNIST, FEMNIST, Shakespeare, and Sent140 under non-IID settings and achieved up to ∼22% higher test accuracy than FedAvg in highly non-IID and straggler-heavy scenarios (for example, 90% stragglers). On non-IID Fashion-MNIST with a CNN (5 clients, 10 rounds), FedProx outperformed FedAvg, FedSGD, and SCAFFOLD, with 91.59% accuracy against 89.95%, 79.43%, and 89.98%, respectively (Dai, 2024). FedProx still transmits model updates without explicit privacy protection, so it remains open to inference-based privacy attacks like FedAvg.

2.2. Differential-Privacy-Based FL

DP-SGD (Fernandez-de Retana, Zulaika, Sánchez-Corcuera and Almeida, 2025) addresses gradient reconstruction attacks: on MNIST and Food-101 CNN experiments, non-DP training allows near-perfect inversion (SSIM ≈ 1), whereas DP-SGD produces noise-like reconstructions, while PDP-SGD fails to give such protection. Bayesian DP (Nguyen, Lai, Tran, Phan and Thai, 2023; Triastcyn and Faltings, 2019) gives tighter ϵ-bounds against inference attacks such as MIAs ($\epsilon \approx 1$ for client-level and ≈ 0.1 for instance-level privacy), but stronger privacy costs utility: standard DP can drop from 97% to 12% accuracy on non-IID MNIST under strong protection, and even Bayesian DP retains only 62%. Adap DP-FL (Fu, Chen and Han, 2022) uses adaptive gradient clipping and noise scheduling and, under $\epsilon = 2$ and $\delta = 10^{-5}$, reaches 94.15% on MNIST and 79.56% on Fashion-MNIST. An adaptive DP framework for federated survival analysis (Niu, He, Zhou, Li, Jiang, Li and Chen, 2025) allocates privacy layer-sensitively and improves the C-index from 0.5843–0.6508 to 0.6276–0.6698 under $\epsilon = 1$–3. LDP-Fed (Truex, Liu, Chow, Gursoy and Wei, 2020) moves to local DP, where clients perturb updates before transmission; with top-k selection it achieves up to

95.21% on MNIST and 82.20% on Fashion-MNIST and reduces communication cost by up to 95%, yet strong privacy budgets still reduce accuracy, especially with few clients. ALDP-FL (Cui and Wu, 2025) adds layer-wise adaptive clipping and bounded perturbation; with $\epsilon = 4$ and $\delta = 10^{-7}$ it improves accuracy by 3–13%, reduces SSIM by ∼85%, and increases MSE by ∼391% compared with FedAvg.

2.3. Secure Aggregation

Secure aggregation replaces noise injection with pairwise masking so that only aggregated updates are revealed. The protocol of Bonawitz et al. (2017) computes secure sums with about 1.73×–1.98× communication overhead and preserves exact accuracy because masks cancel perfectly, but each client pairs with all others, giving $O(n \cdot m)$ communication cost. GVSA (Zhou, Wang, Chen, Wang and Yuan, 2025) restricts communication to groups (for example, 9 peers in a 10-user group), reaching only ∼7% overhead over FedAvg and 2.3× faster execution than earlier secure aggregation, with near-identical accuracy (0.888 versus 0.885 on MNIST). These masking-based methods do not fully prevent membership inference or reconstruction-style attacks, since aggregated updates can leak information through repeated observations across rounds.

2.4. Synthesis of Research Gaps

FedProx stabilizes non-IID training but offers no privacy. DP-based methods protect against reconstruction and membership inference but trade accuracy for privacy, and masking-based secure aggregation preserves accuracy but requires large-scale pairwise communication and does not fully prevent leakage from aggregated updates. FPMANR addresses these limitations by combining FedProx-based local training, layer-wise adaptive noise scaled to update magnitude, and two-neighbor pairwise masking in a ring. Unlike masking methods that use all-pairs or group-based communication (Bonawitz et al., 2017; Zhou et al., 2025), FPMANR uses only two neighbors, reducing communication complexity. Unlike noise-only local DP methods (Niu et al., 2025; Truex et al., 2020), it applies noise in proportion to each layer's update magnitude to limit leakage while preserving utility. Table 8 compares these design characteristics in detail.

3. Proposed Methodology

This section presents the entities and the stages of FPMANR. Fig. 1 illustrates the overall architecture and Algorithm 1 summarizes one communication round.

3.1. Involved Entities

The framework consists of a central server and multiple clients. The **server** initializes the global model and coordinates training. In each round it selects participating clients, distributes the current global model θ_G, receives the masked client updates, aggregates them into the average update, and updates the global model. It also evaluates the model on training, validation, and test data. Each **client** stores a private local dataset d_i following a non-IID distribution. When selected, a client receives θ_G, trains locally with FedProx, and computes its update Δ_i. Before transmission it applies adaptive noise and pairwise masking with two neighboring clients. Clients keep their raw data and original updates private and transmit only masked updates.

3.2. Local Update Computation

At the start of training the server initializes the global model with random parameters θ_G. In each round it randomly selects 30 of the 90 training clients and sends them θ_G. Each selected client C_j trains on its local dataset d_j with FedProx, which stabilizes training under non-IID settings by restricting the deviation of the local model from the global model. The local loss is

$$L_j = \frac{1}{B} \sum_{(x,y)\in\text{batch}} \text{CrossEntropy}\left(\text{model}(x), y\right) + \frac{\mu}{2} \sum_{k=1}^{K} \left\| \theta_j^k - \theta_G^k \right\|^2, \tag{1}$$

where x is the input, y its label, B the batch size, μ the proximal coefficient, $k = 1 \ldots K$ indexes layers or parameter tensors, θ_j^k are client C_j's parameters for layer k, and θ_G^k the corresponding global parameters. The first term is the cross-entropy loss and the second penalizes deviation of the local parameters from the global model. After local training, each client computes its update

$$\Delta_j = \theta_j - \theta_G, \tag{2}$$

where θ_j is the locally trained model of C_j and Δ_j is the change of all model parameters caused by local training.

3.3. Layer-Wise Adaptive Gaussian Noise

If a client transmitted its raw update Δ_j, the server could infer sensitive information such as dominant classes or approximate input patterns. Masking conceals individual updates in transmission, but Δ_j may still carry information about local data; for instance, large parameter values usually indicate that the client has many samples of a certain class or pattern. Small adaptive Gaussian noise is therefore added to each layer of the local update before any masking. The noise magnitude is controlled by the adaptive noise factor (ANF) α. For client C_j and layer l,

$$\Delta_j^{l,\text{noisy}} = \Delta_j^l + n_j^l, \qquad n_j^l \sim \mathcal{N}\left(0, \sigma_l^2\right), \qquad \sigma_l = \alpha \left\| \Delta_j^l \right\|_2, \tag{3}$$

where n_j^l is a zero-mean Gaussian noise vector added element-wise to all parameters of layer l, $\alpha > 0$ controls the noise magnitude, and $\alpha = 0$ means no noise. The noisy update Δ_j^{noisy} combines the noisy updates of all layers. Each client temporarily stores Δ_j^{noisy} until the server confirms successful aggregation or triggers recovery.

Because the noise scales with the norm of each layer update, layers with larger updates, which may reveal more about local data, receive proportionally larger noise, while

Table 1
Summary of representative studies on non-IID optimization, differential-privacy-based FL, and secure aggregation, with their key reported results and the limitations that FPMANR addresses.

Ref.	Approach	Evaluation setting	Key results	Limitation
McMahan et al. (2017)	FedAvg: local SGD with global averaging	MNIST CNN, Shakespeare	99% MNIST, 54% Shakespeare; ≈18–65 rounds (IID) versus 173–600 (non-IID) on MNIST	Slow convergence from client drift under non-IID data
Li et al. (2020)	FedProx: proximal term, partial local work	Synthetic, MNIST, FEMNIST, Shakespeare, Sent140	Up to ∼22% higher test accuracy than FedAvg in highly non-IID, straggler-heavy cases	No privacy protection
Dai (2024)	Comparison of FedProx, FedAvg, FedSGD, SCAFFOLD	Non-IID Fashion-MNIST, 5 clients, 10 rounds	FedProx 91.59% (FedAvg 89.95%, SCAFFOLD 89.98%, FedSGD 79.43%)	Plain updates remain open to inference attacks
Fernandez-de Retana et al. (2025)	DP-SGD against gradient inversion	MNIST, Food-101 CNNs	Non-DP: SSIM ≈ 1; DP-SGD: noise-like reconstructions	Noise-based protection reduces utility
Nguyen et al. (2023); Triastcyn and Faltings (2019)	Bayesian DP, tighter ϵ-bounds	Non-IID MNIST	$\epsilon \approx 1$ (client), ≈ 0.1 (instance); standard DP 97%→12%, Bayesian DP 62%	Strong privacy costs accuracy
Fu et al. (2022)	Adap DP-FL: adaptive clipping and noise scheduling	MNIST, Fashion-MNIST ($\epsilon = 2$, $\delta = 10^{-5}$)	94.15% (MNIST), 79.56% (Fashion-MNIST)	Privacy–utility trade-off remains
Niu et al. (2025)	Layer-sensitive adaptive DP allocation	Federated survival analysis, $\epsilon = 1$–3	C-index 0.5843–0.6508 → 0.6276–0.6698	Targets survival analysis, noise-only
Truex et al. (2020)	LDP-Fed: local DP, top-k selection	MNIST, Fashion-MNIST	Up to 95.21% and 82.20%; up to 95% less communication	Strong budgets reduce accuracy with few clients
Cui and Wu (2025)	ALDP-FL: layer-wise adaptive clipping, bounded noise	Local DP, $\epsilon = 4$, $\delta = 10^{-7}$	Accuracy +3–13%, SSIM − ∼85%, MSE +∼391% versus FedAvg	Relies on noise injection alone
Bonawitz et al. (2017)	Secure aggregation with pairwise masking	Secure sum computation	1.73×–1.98× overhead; exact accuracy	$O(n \cdot m)$ cost from all-pairs masking; leakage from aggregates
Zhou et al. (2025)	GVSA: group-based masking and secret sharing	MNIST	∼7% overhead over FedAvg; 2.3× faster; 0.888 versus 0.885	Leakage from aggregates over repeated rounds

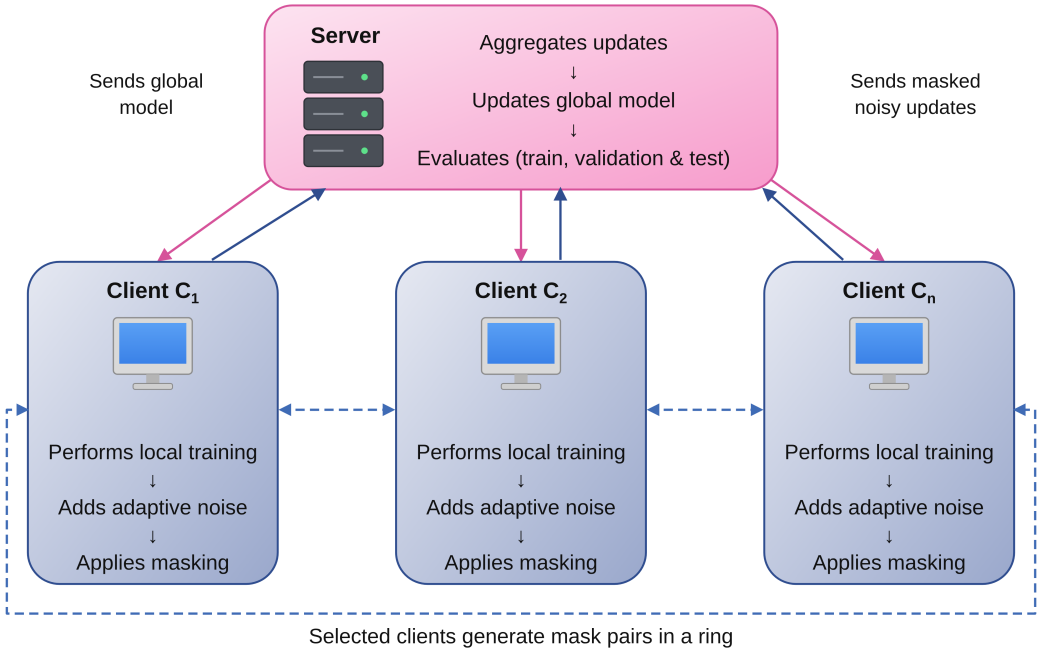


Figure 1: Architecture of the proposed FPMANR framework. The server selects clients, sends the global model, aggregates the received updates, updates the global model, and evaluates it on training, validation, and test data. Each selected client (C_1, C_2, ..., C_n) performs local training with FedProx, adds adaptive Gaussian noise, and applies pairwise masking. The dashed ring connecting the clients indicates the mask pairs generated between neighboring clients, and the arrows to the server indicate the transmission of masked noisy updates.

layers with smaller updates receive less. This improves privacy while limiting unnecessary performance loss, in contrast to conventional DP methods that apply uniform noise to all layers. We study $\alpha = 0.1$, a mild perturbation that reduces leakage while keeping accuracy high, and $\alpha = 0.2$, a stronger perturbation that further improves privacy at a larger accuracy cost. Increasing α therefore strengthens privacy at the cost of accuracy and training stability. The approach reduces information leakage but does not provide a formal differential privacy guarantee.

3.4. Ring-Based Pairwise Masking

In each round the server builds a ring only among the selected clients, giving each participating client C_j two neighbors, C_i (left) and C_k (right). Each pair of neighbors shares one random mask vector. The mask shared by C_j and its right neighbor C_k is denoted s_{jk} (with $s_{jk} = s_{kj}$), and similarly s_{ij} for the left neighbor. Client C_j first scales its noisy update by its aggregation weight $w_j = n_j / \sum_{k \in S} n_k$, where n_j is its number of local samples and S the set of active clients, and then forms its masked update by adding the mask shared with its right neighbor and subtracting the mask shared with its left neighbor:

$$\widetilde{\Delta}_j = w_j \, \Delta_j^{\text{noisy}} - s_{ij} + s_{jk}. \tag{4}$$

Applying the weights on the client keeps the masks cancelling when sample counts differ. The resulting noisy and masked update $\widetilde{\Delta}_j$ is sent to the server, while individual updates remain private. Because every mask appears once with a plus sign (at one neighbor) and once with a minus sign (at the other), the masks cancel over the whole ring:

$$\sum_{j \in S} \left(-s_{ij} + s_{jk} \right) = 0. \tag{5}$$

3.5. Client Dropout Handling and Recovery

In practical FL systems, selected clients may become unavailable because of communication failures or device interruptions, which prevents mask cancellation. The recovery mechanism restores correct aggregation using the temporarily stored noisy update Δ_j^{noisy} and requires no local retraining. Table 2 describes the four handled cases. In all cases the affected active clients discard their previously computed masked updates, remove the masks associated with failed clients, establish a new pairwise connection with the nearest active client (generating a new shared mask $s_{ik} = s_{ki}$), apply the new mask together with their remaining valid masks to their original noisy updates, and resend the repaired updates. Aggregation then proceeds over the active clients that successfully transmit. If additional failures occur during recovery, the server repeats the repair; if the number of active clients falls below the predefined threshold of 20, the round is aborted and a new client selection is started. Recovery takes place within the current round and does not affect completed rounds.

3.6. Global Model Aggregation and Update

The server sums the received masked updates, which equals the weighted average of the noisy updates because

Table 2
Client dropout cases handled by the recovery mechanism, with the failure point, its effect on mask cancellation, and the repair action. C_h, C_i, C_j, C_k, C_l denote consecutive clients in the ring.

Case	Failure point	Effect	Repair action
1	C_j fails before pairwise masks are established	Neighbors C_i and C_k detect the missing connection and notify the server	The server removes C_j; C_i and C_k establish a direct connection and generate a new mask s_{ik}; training continues normally
2	C_j fails after mask generation but before transmitting its update	The masks of C_i and C_k tied to C_j cannot cancel; the server discards their updates	C_i and C_k drop the masks tied to C_j, connect directly with a new mask, rebuild from their stored noisy updates, and resend
3	C_j transmitted its masked update and then a neighbor (C_i or C_k) fails before transmitting	C_j is no longer available to re-pair; the server discards C_j's update and the affected neighboring updates	If C_i fails, C_h and C_k pair directly; if C_k fails, C_i and C_l pair directly; affected active clients rebuild and resend
4	Consecutive clients $C_p, \ldots, C_q$ become inactive	Masks tied to the inactive clients cannot cancel; affected received updates are discarded	The nearest active clients C_{p-1} and C_{q+1} pair directly with a new mask; affected clients rebuild and resend

Algorithm 1 One FPMANR communication round

1: **Server:** randomly select 30 of the 90 training clients; build the ring; send θ_G.
2: **for** each selected client C_j **in parallel do**
3: Train with FedProx (Eq. 1); compute $\Delta_j = \theta_j - \theta_G$ (Eq. 2).
4: Add layer-wise adaptive noise (Eq. 3); store Δ_j^{noisy}.
5: Form $\widetilde{\Delta}_j$ with the two neighbor masks (Eq. 4) and send it to the server.
6: **end for**
7: **Server:** if clients drop out, trigger the repair of Table 2; abort the round if fewer than 20 clients remain active.
8: **Server:** aggregate (Eq. 6); update θ_G (Eq. 7); evaluate on training, validation, and test data.

the masks cancel (Eq. 5):

$$\Delta_{avg} = \sum_{j \in S} \widetilde{\Delta}_j = \sum_{j \in S} \frac{n_j}{\sum_{k \in S} n_k} \Delta_j^{\text{noisy}}. \tag{6}$$

Thus the server observes only the weighted aggregation of the clients' noisy updates, while individual updates remain private. The global model is then updated as

$$\theta_G^{new} = \theta_G^{old} + \Delta_{avg}. \tag{7}$$

After each round, the updated global model is evaluated with cross-entropy loss and classification accuracy on the training data of the selected clients, on the data of the 10 fixed validation clients, and on the test set. The server then randomly selects 30 new clients for the next round, and training continues until completion. Algorithm 1 summarizes the full round.

3.7. Threat Model

FPMANR is evaluated under a standard FL threat model. The server is honest-but-curious: it follows the training and aggregation procedures but may try to infer information from received updates. Clients follow the protocol, performing local training, adaptive noise addition, and masking as specified, and are assumed not to share masked updates with one another; only the server receives transmitted updates. Dropouts may occur and are handled by the recovery mechanism (Section 3.5). Each round is assumed to include at least twenty participating clients ($|S| \geq 20$), so that the server observes aggregated updates rather than any individual contribution.

Honest-but-curious server. The server receives only noisy masked updates $\widetilde{\Delta}_j$. After aggregation it obtains only the weighted sum of noisy updates (Eq. 6). During recovery the server only coordinates the repair, and the stored noisy updates remain on the clients. Recovering an individual update from the aggregate is mathematically underdetermined, because many combinations of client updates yield the same aggregate, so exact reconstruction of local client data is considerably harder.

Gradient reconstruction attacks. Attackers may try to extract dominant classes or hidden patterns, or reconstruct individual gradients, with DLG, iDLG, and GradInv. Since each client first perturbs its layer updates and then masks them, and repaired updates are rebuilt from stored noisy updates without retraining, identifying client patterns or reconstructing local gradients becomes substantially more difficult.

Client collusion. Colluding clients may share their pairwise masks. Because clients do not have access to other clients' transmitted updates, collusion only allows removal of shared mask components, and the adaptive noise remains in the underlying update. The recovery mechanism uses only locally stored noisy updates, so dropouts expose no additional information.

Non-IID leakage, membership inference, model inversion, and label leakage. Dirichlet partitioning ($\beta = 0.1$) leaves each client with a skewed subset of classes, which may expose class-specific information through updates. FPMANR combines FedProx with adaptive noise, and because

the server only obtains the aggregated noisy result after mask cancellation, it is considerably harder to determine whether a sample belongs to a client dataset, to reconstruct client data, or to infer class labels.

Scope and limitations. Fully malicious clients that manipulate updates, transmit incorrect masks, collaborate with the server, or provide incorrect information during recovery are out of scope. Some residual privacy risk may remain in extreme cases, for example when client datasets are extremely small, highly unique, or severely skewed in class distribution.

4. Experiments

This section describes the datasets, client distribution, model, experimental environment, compared settings, and evaluation metrics.

4.1. Dataset Description

Experiments use three benchmark image classification datasets: EMNIST Digits (Cohen, Afshar, Tapson and Van Schaik, 2017), EMNIST Letters (Cohen et al., 2017), and Fashion-MNIST (Xiao, Rasul and Vollgraf, 2017). All contain 28×28 grayscale images and are widely used to evaluate ML and FL models. The training sets are distributed among clients and the test sets evaluate the global model; the task is multi-class image classification. Table 3 lists the dataset statistics. EMNIST Digits contains handwritten digits 0–9 from different writers, whose handwriting variation makes the benchmark challenging. EMNIST Letters contains 26 balanced classes of handwritten letters and is harder than digits because several letters look alike. Fashion-MNIST contains fashion products such as shirts, trousers, shoes, bags, and coats, with more detailed visual patterns and greater inter-class similarity than the handwritten datasets.

Table 3: Statistics of the three benchmark datasets used in the experiments: number of training images, test images, and classes. All images are 28×28 grayscale.

Dataset	Train	Test	Classes
EMNIST Digits (Cohen et al., 2017)	240,000	40,000	10
EMNIST Letters (Cohen et al., 2017)	124,800	20,800	26
Fashion-MNIST (Xiao et al., 2017)	60,000	10,000	10

4.2. Dataset and Client Distribution

The training data are distributed among 100 clients with a Dirichlet distribution of concentration parameter $\beta = 0.1$, producing highly skewed, imbalanced, and heterogeneous local datasets in which each client holds samples from only a subset of classes (a strict non-IID setting). Formally, the dataset D is partitioned into $\{d_1, d_2, \ldots, d_{100}\}$, and each client j privately keeps d_j throughout training. Ten fixed clients are reserved for validation and do not take part in local training. In each round, 30 clients (30% of all clients) are randomly selected from the remaining 90 training clients, and the rest stay inactive. Table 4 summarizes the federated configuration.

4.3. Model Architecture and Configuration

A unified CNN is used by all clients and by the server for all datasets. It has two convolutional layers with 3×3 kernels and 32 and 64 feature maps, each followed by ReLU and 2×2 max-pooling, then a fully connected layer with 256 ReLU units, and a final output layer that produces logits (10 classes for EMNIST Digits and Fashion-MNIST, 26 for EMNIST Letters). The compared settings are FP (FedProx), FPM (FedProx with pairwise masking), FPMANR 0.1 (masking with $\alpha = 0.1$), and FPMANR 0.2 (masking with $\alpha = 0.2$). All settings use the same architecture, data distribution, and training configuration for 60 communication rounds (Table 4).

Table 4: Federated learning configuration used in all experiments.

Parameter	Value
Total clients	100
Training clients / validation clients	90 / 10
Clients selected per round	30
Data partition	Dirichlet, $\beta = 0.1$
Communication rounds	60
Adaptive noise factor α	0.1, 0.2
Minimum active clients (recovery threshold)	20
Local objective	FedProx, Eq. 1
Compared settings	FP, FPM, FPMANR 0.1, FPMANR 0.2

4.4. Experimental Environment

The experiments were performed on Google Colab with an NVIDIA T4 GPU, and the implementation was developed in a Jupyter Notebook (.ipynb). FPMANR was simulated on a single machine, so real network communication delays between clients and server were not considered.

4.5. Evaluation Metrics

Model utility is evaluated with training, validation, and test accuracy and cross-entropy loss, and with total training time. Reconstruction quality under DLG, iDLG, and GradInv is measured with MSE, PSNR, SSIM, and LPIPS. Higher MSE and LPIPS and lower PSNR and SSIM indicate stronger protection. The attack success rate (ASR) is the percentage of reconstructed samples with SSIM ≥ 0.1 or PSNR ≥ 10, so lower ASR means fewer recognizable reconstructions. PSNR, SSIM, and LPIPS measure visual, structural, and perceptual similarity (Carletti, Foggia, Mazzocca, Parrella and Vento, 2026), MSE measures pixel-level error (Shi, Kotevska, Reshniak, Singh and Raskar, 2024), and ASR quantifies successful reconstructions under predefined quality criteria (He, Stevens, Ouyang, De Smedt, Barros and Moreira, 2025). The attacks are described in Zhang, Guo, Wang, Xie and Tao (2022). DLG reconstructs data and labels by minimizing the gradient difference, iDLG infers true labels directly from gradients in single-batch settings, and GradInv uses total variation and regularization terms. Membership inference is evaluated with confidence-based, loss-based, and entropy-based attacks (Shokri, Stronati, Song

Table 5
Final-round (round 60) training, validation, and test accuracy (%) and test cross-entropy loss of FP, FPM, FPMANR 0.1, and FPMANR 0.2 on the three datasets. The last column gives the difference in test accuracy relative to FP in percentage points (pp).

Dataset	Setting	Train acc.	Val. acc.	Test acc.	Test loss	Change (pp)
EMNIST Digits	FP	98.68	98.01	97.92	0.0668	–
	FPM	98.76	98.40	98.52	0.0484	+0.60
	FPMANR 0.1	98.56	98.02	98.10	0.0606	+0.18
	FPMANR 0.2	97.69	96.35	96.57	0.1086	−1.35
EMNIST Letters	FP	92.79	87.90	90.53	0.2837	–
	FPM	92.17	88.33	90.45	0.2945	−0.08
	FPMANR 0.1	87.60	82.94	85.39	0.4485	−5.14
	FPMANR 0.2	74.74	74.16	73.37	0.9043	−17.16
Fashion-MNIST	FP	82.91	77.95	80.57	0.5166	–
	FPM	73.23	67.99	73.71	0.7403	−6.86
	FPMANR 0.1	74.31	68.72	74.51	0.7086	−6.06
	FPMANR 0.2	73.23	68.74	73.12	0.8328	−7.45

and Shmatikov, 2017; Banerjee, Roy, Ahamed, Quinn, Vucovich, Nandakumar, Choi, Rahman, Bowen and Shetty, 2023), reporting the corresponding AUC values and attack accuracy; an AUC of 0.5 and an accuracy of 50% indicate random guessing, and lower values indicate stronger protection. Dropout recovery is evaluated with the residual mask norm $R = \left\| \sum_{i \in S} M_i \right\|$, where M_i is the masking vector of active client i and S is the set of active clients in the round; a value close to zero indicates successful mask cancellation.

5. Results

5.1. Model Utility

Table 5 and Fig. 2 compare the four settings over 60 communication rounds. After 60 rounds, FP reaches 97.92% test accuracy on EMNIST Digits, 90.53% on EMNIST Letters, and 80.57% on Fashion-MNIST. Introducing pairwise masking (FPM) largely maintains performance on the EMNIST datasets, with test accuracy changes of +0.60 and −0.08 percentage points (pp). Ideally FP and FPM give identical results since the masks cancel during aggregation; small differences arise from floating-point precision, because the masking step adds computations and rounding errors accumulate when many updates are summed, behaving like very small noise on the true update. On EMNIST the updates are stable and vary little between clients, so this numerical noise is negligible. On Fashion-MNIST the updates vary more across clients because of higher data complexity and less uniform class structure, so the same effects have a larger impact on the aggregated update direction, especially in early rounds, which produces a visible gap (a test-accuracy change of −6.86 pp). The gap does not grow over time because the per-round perturbations are independent and small relative to the update scale.

Training time remains comparable after adding masking. On EMNIST Digits, FP takes 4949 s and FPM 4845 s (−2.10%); on EMNIST Letters, 2589 s and 2650 s (+2.36%); on Fashion-MNIST, 1229 s and 1302 s (+5.94%). Pairwise masking therefore adds only a small computational overhead.

Adaptive noise reduces accuracy slightly further, as expected. Relative to FP, FPMANR 0.1 changes the test accuracy by +0.18, −5.14, and −6.06 pp on EMNIST Digits, EMNIST Letters, and Fashion-MNIST, while FPMANR 0.2 changes it by −1.35, −17.16, and −7.45 pp. These results show the expected privacy–utility trade-off, in which stronger adaptive noise improves privacy at the cost of model performance.

5.2. Gradient Reconstruction Attack Analysis

Fig. 3 shows the visual results of gradient inversion on EMNIST Digits (a), Fashion-MNIST (b), and EMNIST Letters (c), and Table 6 reports the quantitative results. Under FP, all three attacks recover recognizable image structures, with GradInv giving the clearest reconstructions because of its stronger optimization and regularization. The recovered digits, letters, and clothing shapes show that substantial input information remains in the transmitted updates. With pairwise masking (FPM), reconstruction quality decreases visually, because random masks perturb each client update, and partial low-level structures remain observable in some samples. With adaptive noise (FPMANR 0.1 and 0.2), the reconstructions become increasingly distorted and unrecognizable. Since the noise scales with each layer's update magnitude, the perturbed updates differ more from the original updates, which makes gradient matching considerably harder, and a larger ANF lowers the signal-to-noise ratio further.

Quantitatively (Table 6), FP and FPM generally show lower MSE and LPIPS and higher PSNR, SSIM, and ASR, so substantial information can still be recovered. FPM shows slightly higher ASR than FP in some cases (for example, 11.54% versus 7.69% for DLG on EMNIST Letters), because masking perturbs updates without destroying the underlying gradient structure. After adaptive noise is introduced, MSE and LPIPS increase while PSNR, SSIM, and ASR

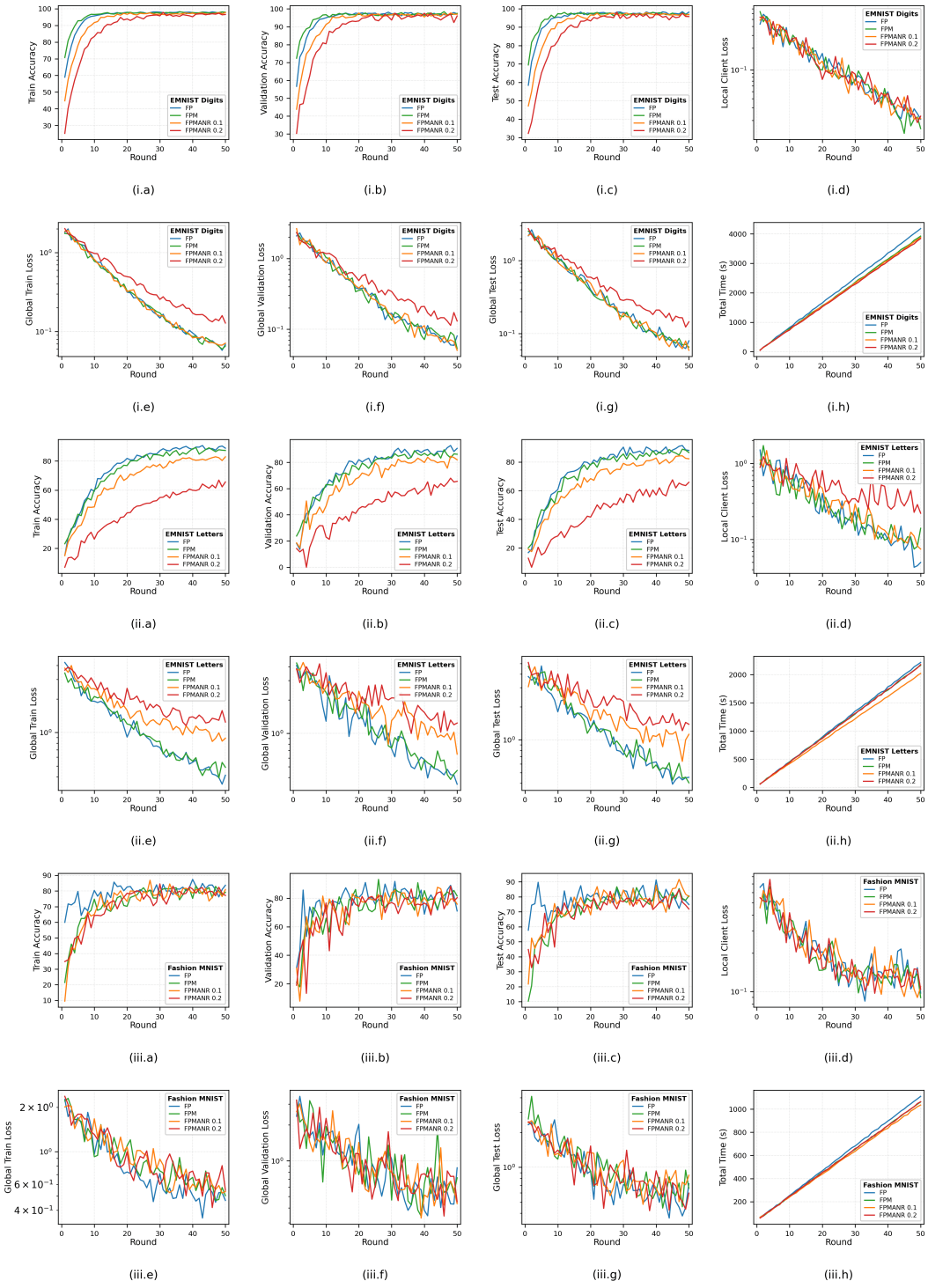
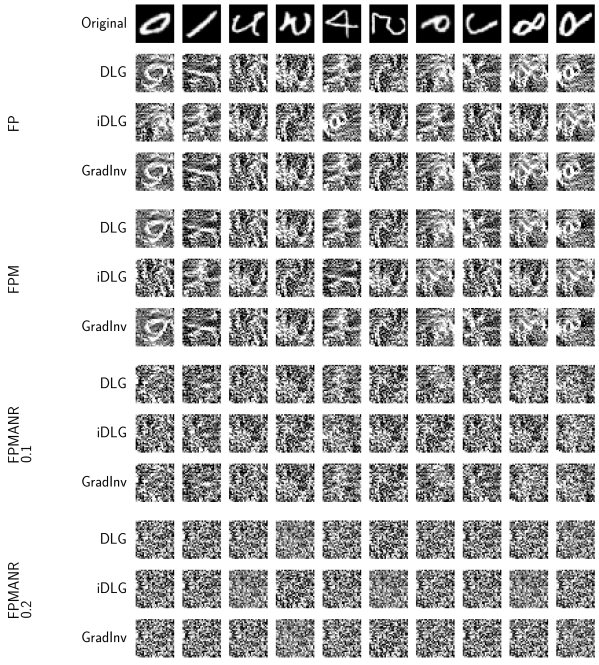
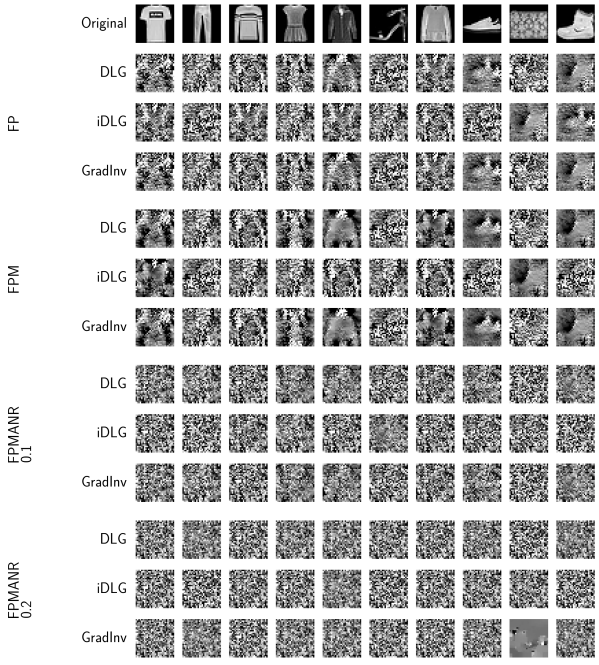


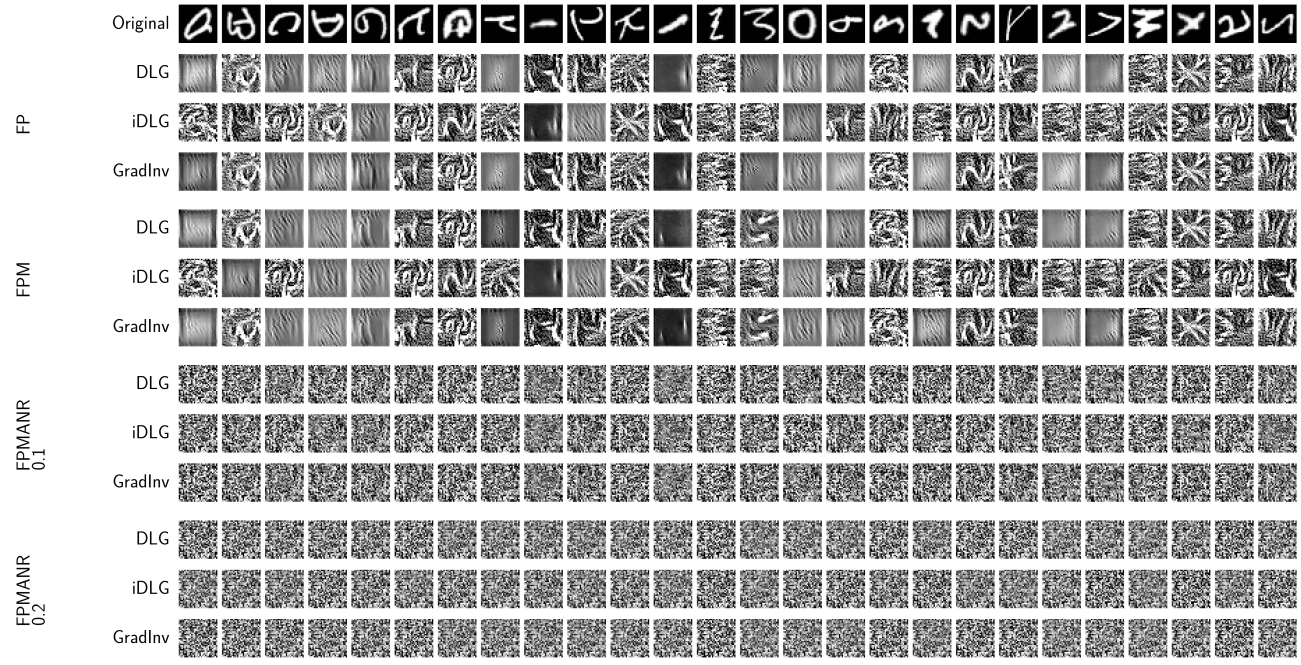
Figure 2: Accuracy, loss, and total training time over the communication rounds for FP, FPM, FPMANR 0.1, and FPMANR 0.2 on (i) EMNIST Digits, (ii) EMNIST Letters, and (iii) Fashion-MNIST. For each dataset, panels (a)–(c) show the global train, validation, and test accuracy, panel (d) the local client loss, panels (e)–(g) the global train, validation, and test loss (logarithmic scale), and panel (h) the cumulative training time in seconds.



(a) EMNIST Digits



(b) Fashion-MNIST



(c) EMNIST Letters

Figure 3: Gradient reconstruction results under FP, FPM, FPMANR 0.1, and FPMANR 0.2 on (a) EMNIST Digits, (b) Fashion-MNIST, and (c) EMNIST Letters. In each panel the top row shows the original training samples; for each setting, the three rows show the reconstructions produced by DLG, iDLG, and GradInv. Recognizable structure remains under FP and FPM, while FPMANR 0.1 and FPMANR 0.2 give increasingly distorted reconstructions.

Table 6
Quantitative evaluation of gradient reconstruction attacks (DLG, iDLG, GradInv) under FP, FPM, FPMANR 0.1, and FPMANR 0.2 on EMNIST Digits, EMNIST Letters, and Fashion-MNIST. ↑ (↓) indicates that a higher (lower) value means stronger protection. ASR is the percentage of reconstructions with SSIM ≥ 0.1 or PSNR ≥ 10.

Dataset	Setting	Attack	MSE ↑	PSNR ↓	SSIM ↓	LPIPS ↑	ASR ↓
EMNIST Digits	FP	DLG	0.3151	5.0220	0.0464	0.6201	10.00%
		iDLG	0.3333	4.7737	0.0195	0.6268	0.00%
		GradInv	0.3149	5.0245	0.0467	0.6200	10.00%
	FPM	DLG	0.3115	5.0768	0.0514	0.6350	10.00%
		iDLG	0.3243	4.9051	0.0360	0.6348	0.00%
		GradInv	0.3114	5.0781	0.0514	0.6349	10.00%
	FPMANR 0.1	DLG	0.3201	4.9492	0.0115	0.6486	0.00%
		iDLG	0.3300	4.8163	0.0167	0.6538	0.00%
		GradInv	0.3200	4.9498	0.0115	0.6486	0.00%
	FPMANR 0.2	DLG	0.3294	4.8242	0.0121	0.6410	0.00%
		iDLG	0.3275	4.8507	0.0085	0.6410	0.00%
		GradInv	0.3287	4.8332	0.0124	0.6411	0.00%
EMNIST Letters	FP	DLG	0.2653	5.8435	0.0374	0.5661	7.69%
		iDLG	0.2921	5.4771	0.0116	0.5542	0.00%
		GradInv	0.2625	5.9408	0.0384	0.5617	3.85%
	FPM	DLG	0.2539	6.0800	0.0395	0.5657	11.54%
		iDLG	0.2831	5.6363	0.0222	0.5540	7.69%
		GradInv	0.2559	6.0545	0.0407	0.5648	11.54%
	FPMANR 0.1	DLG	0.3098	5.0982	0.0067	0.6529	0.00%
		iDLG	0.3140	5.0388	0.0053	0.6487	0.00%
		GradInv	0.3098	5.0985	0.0067	0.6529	0.00%
	FPMANR 0.2	DLG	0.3147	5.0294	0.0023	0.6454	0.00%
		iDLG	0.3135	5.0472	0.0010	0.6465	0.00%
		GradInv	0.3146	5.0304	0.0024	0.6461	0.00%
Fashion-MNIST	FP	DLG	0.2112	6.8415	0.0771	0.4809	40.00%
		iDLG	0.2294	6.5660	0.0365	0.4950	10.00%
		GradInv	0.2111	6.8438	0.0771	0.4808	40.00%
	FPM	DLG	0.1857	7.4734	0.1078	0.4251	40.00%
		iDLG	0.2365	6.4609	0.0356	0.4852	30.00%
		GradInv	0.1856	7.4744	0.1078	0.4254	40.00%
	FPMANR 0.1	DLG	0.2244	6.5520	0.0216	0.5102	0.00%
		iDLG	0.2491	6.0995	−0.0011	0.5173	0.00%
		GradInv	0.2242	6.5555	0.0216	0.5102	0.00%
	FPMANR 0.2	DLG	0.2321	6.3856	0.0063	0.5235	0.00%
		iDLG	0.2521	6.0645	−0.0076	0.5055	0.00%
		GradInv	0.2206	6.7219	0.0082	0.5562	10.00%

decrease. With FPMANR 0.1, SSIM for DLG falls from 0.0464 to 0.0115 on EMNIST Digits, from 0.0374 to 0.0067 on EMNIST Letters, and from 0.0771 to 0.0216 on Fashion-MNIST, and ASR is 0% in all nine dataset–attack combinations, compared with up to 40% for FP on Fashion-MNIST. FPMANR 0.2 provides the strongest protection in most cases. The exception is Fashion-MNIST with GradInv, where ASR is 10% under FPMANR 0.2 against 0% under FPMANR 0.1; the corresponding SSIM is only 0.0082 and LPIPS reaches its highest value of 0.5562, so the reconstructions are still highly distorted. This occurs because ASR is based on a small number of recognizable samples, so small variations appear between closely related noise settings even while overall reconstruction quality keeps degrading.

5.3. Membership Inference Attack Analysis

Fig. 4 and Table 7 report the confidence-, loss-, and entropy-based MIAs. Across all datasets and settings, AUC values remain very close to 0.5 and attack accuracies stay near 50%, including for the unprotected FP baseline, indicating that the attacker cannot reliably distinguish members from non-members. For EMNIST Digits, FP and FPM give AUC values around 0.49 with attack accuracies between 49.60% and 50.15%, and adaptive noise moves the AUC values slightly closer to 0.5. On EMNIST Letters, FP and FPM reach AUC values slightly above 0.50, whereas FPMANR 0.1 and 0.2 reduce the confidence- and entropy-based AUC toward 0.49–0.50. On Fashion-MNIST, FP and FPM show the highest AUC values of the three datasets (about 0.51–0.52), which are still very close to 0.5; adaptive noise reduces the confidence- and entropy-based AUC further, with FPMANR 0.2 closest to the random baseline. The ROC curves in Fig. 4 and the values in Table 7 consistently show that MIAs are largely ineffective in all evaluated settings.

5.4. Dropout Recovery Validation

Fig. 5 evaluates whether the recovery mechanism preserves mask cancellation under client dropouts using the residual mask norm R. With recovery, R stays near zero in all rounds even when clients drop out. Without recovery, nonzero residual masks appear whenever clients drop out, and the residual returns to near zero only in rounds without dropouts. These results confirm that the recovery mechanism preserves correct mask cancellation and aggregation consistency under varying client participation.

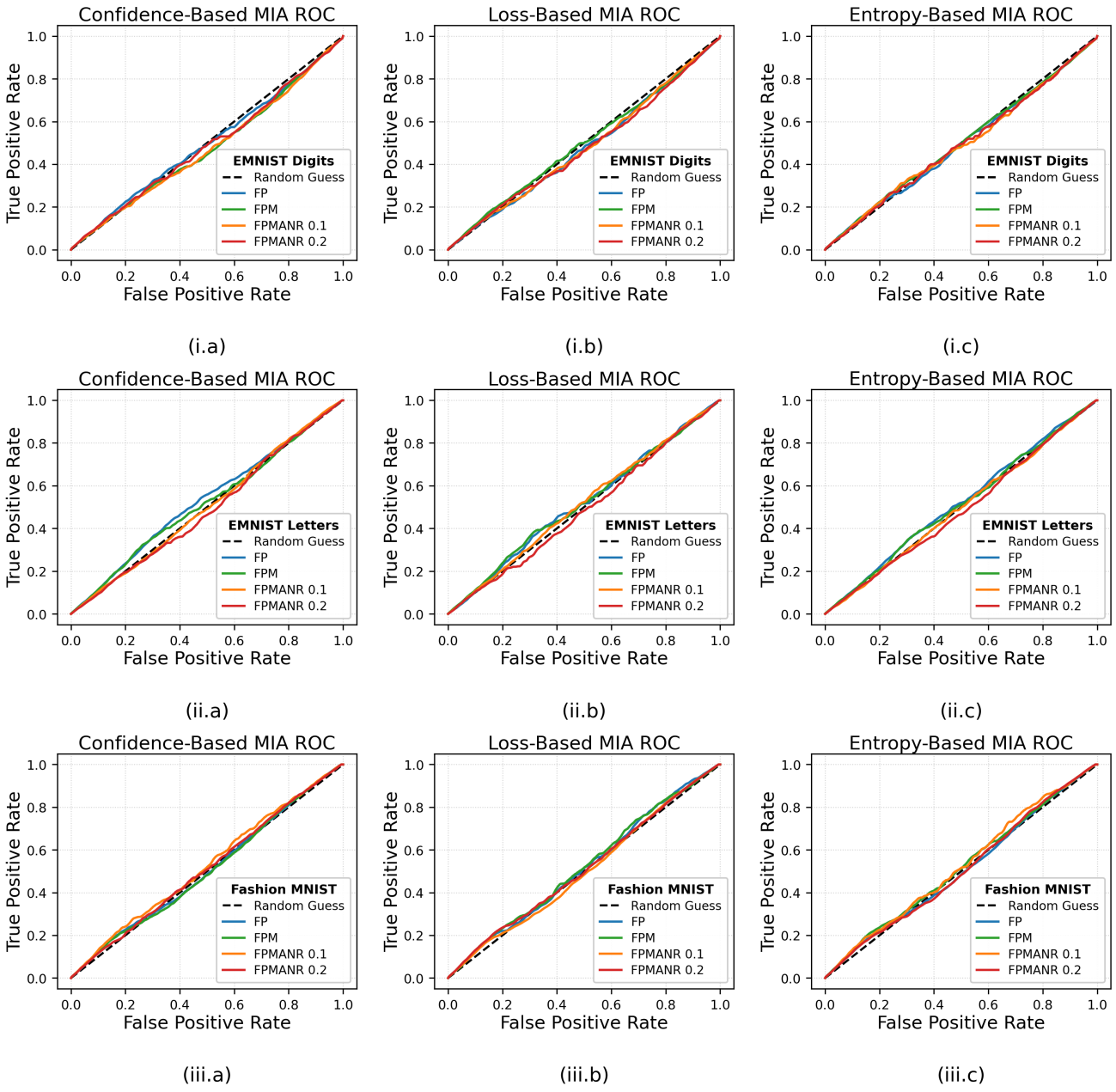


Figure 4: ROC curves of membership inference attacks under FP, FPM, FPMANR 0.1, and FPMANR 0.2 on (i) EMNIST Digits, (ii) EMNIST Letters, and (iii) Fashion-MNIST. For each dataset, panels (a)–(c) show the confidence-based, loss-based, and entropy-based attacks. The dashed diagonal marks random guessing.

Table 7
Membership inference attack performance for FP, FPM, FPMANR 0.1, and FPMANR 0.2 on the three datasets. AUC values near 0.5 and attack accuracy near 50% indicate random guessing; lower attacker performance indicates stronger privacy protection.

Dataset	Setting	Confidence AUC	Loss AUC	Entropy AUC	Attack accuracy (%)
EMNIST Digits	FP	0.4889	0.4893	0.4888	49.60
	FPM	0.4891	0.4892	0.4889	50.15
	FPMANR 0.1	0.4943	0.4952	0.4941	50.15
	FPMANR 0.2	0.4951	0.4952	0.4948	50.00
EMNIST Letters	FP	0.5033	0.5057	0.5027	48.60
	FPM	0.5041	0.5059	0.5033	49.20
	FPMANR 0.1	0.4936	0.4955	0.4931	50.05
	FPMANR 0.2	0.4947	0.4858	0.4927	49.45
Fashion-MNIST	FP	0.5119	0.5148	0.5106	50.05
	FPM	0.5126	0.5169	0.5111	50.90
	FPMANR 0.1	0.5085	0.5145	0.5076	51.30
	FPMANR 0.2	0.5004	0.5145	0.4997	49.40

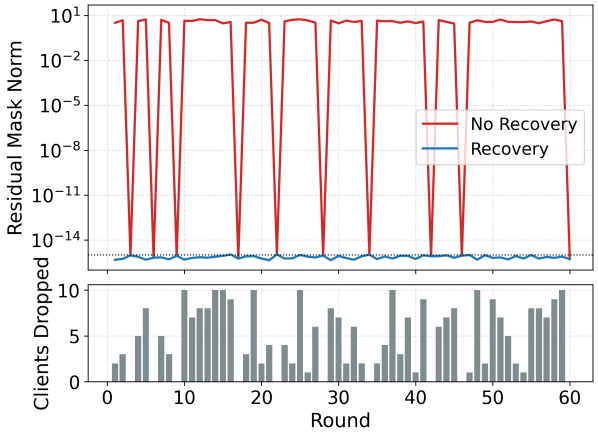


Figure 5: Effect of the recovery mechanism on mask cancellation under client dropouts. The upper panel shows the residual mask norm R (logarithmic scale) per round with and without recovery; the lower panel shows the number of clients dropped in each round.

5.5. Comparison with Existing Methods

Table 8 compares FPMANR with existing FL and privacy-preserving methods in terms of design characteristics. FPMANR integrates adaptive noise and pairwise masking to improve privacy while maintaining efficient communication and stable training under non-IID settings. Table 9 compares the asymptotic computation and communication complexity. FPMANR keeps the computational and communication efficiency of conventional FL while reducing the communication overhead of large-scale masking-based aggregation.

6. Discussion

6.1. Privacy–Utility Trade-off

Considering privacy and utility together, FPMANR 0.1 provides the best balance among the evaluated settings. Compared with FP and FPM, it drives the reconstruction success rate to nearly zero across all datasets (Table 6, Fig. 3) while lowering structural similarity and increasing perceptual distortion, with test-accuracy changes of +0.18, −5.14, and −6.06 pp (Table 5). Membership inference remains close to random guessing, with AUC values near 0.5 and attack accuracies around 50% (Table 7). FPMANR 0.2 offers slightly stronger privacy in some cases, but the additional improvement is small compared with its much larger accuracy loss (up to −17.16 pp on EMNIST Letters). FPMANR 0.1 is therefore selected as the preferred configuration, as it delivers strong resistance to privacy attacks while maintaining high accuracy and training effectiveness.

6.2. Communication and Computational Efficiency

Pairwise masking adds only a small computational overhead (training-time changes of −2.10%, +2.36%, and +5.94% across the datasets), and Table 9 shows that FPMANR retains the per-client cost of conventional FL because each client masks with only two ring neighbors, instead of all participants as in large-scale masking schemes. The recovery mechanism (Fig. 5) keeps aggregation correct under dropouts without local retraining.

6.3. Limitations

FPMANR provides empirical privacy protection and no formal differential privacy guarantee. Fully malicious clients and client–server collusion are outside its scope, and residual privacy risk may remain when client datasets are extremely small, highly unique, or severely skewed in class distribution. The framework was simulated on a single machine, so real network delays were not measured, and the evaluation uses three small-scale image benchmarks. Adaptive noise reduces accuracy, most visibly on EMNIST Letters, and the FP–FPM gap on Fashion-MNIST shows that masking can affect results on more complex data.

6.4. Future Work

Future work includes evaluating FPMANR on larger-scale FL environments and additional real-world datasets, testing it against a wider range of privacy and security attacks, and exploring stronger formal privacy guarantees while further improving the privacy–utility trade-off.

Table 8
Qualitative comparison of FPMANR with existing FL and privacy-preserving methods. ✓ denotes supported and ✗ not supported. "Perturbed" means that aggregation is exact only up to the added noise.

Method	Privacy mechanism	Masking topology	Client drift control	Non-IID support	Dropout handling	Gradient leakage protection	MIA resistance	Exact aggregation
FedAvg (McMahan et al., 2017)	None	Centralized star	✗	Partial	Fractional selection	✗	✗	✓
FedProx (Li et al., 2020)	None	Centralized star	✓ (proximal term)	✓	Partial work tolerated	✗	✗	✓
DP-SGD (Fernandez-de Retana et al., 2025)	Fixed global DP	Centralized star	✗	Partial	Fractional selection	✓	✓	Perturbed
Adap DP-FL (Fu et al., 2022)	Adaptive global DP	Centralized star	✗	Partial	Fractional selection	✓	✓	Perturbed
ALDP-FL (Cui and Wu, 2025)	Adaptive local DP	Centralized star	✗	Partial	Fractional selection	✓	✓	Perturbed
SecAgg (Bonawitz et al., 2017)	Cryptographic SMPC	All-to-all mesh	✗	✓	Fault-tolerant recovery	✓	✗	✓
GVSA (Zhou et al., 2025)	Crypto masking + secret sharing	Group-based	✗	✓	Group secret reconstruction	✓	✗	✓
FPMANR (proposed)	**Adaptive noise + masking**	**Two-neighbor ring**	✓ (proximal term)	✓	**Ring repair + threshold**	✓	✓	Perturbed

Table 9
Asymptotic complexity comparison of FPMANR and existing FL and secure aggregation methods. n: number of active clients; m: number of model parameters; b: mini-batch size; E: number of local epochs; g: group size. †DP-SGD applies clipping and noise at the mini-batch level, so its computation is expressed in the batch size rather than the epoch count.

Method	User computation	User communication	Server communication
FedAvg (McMahan et al., 2017)	$O(E \cdot b \cdot m)$	$O(m)$	$O(n \cdot m)$
FedProx (Li et al., 2020)	$O(E \cdot b \cdot m)$	$O(m)$	$O(n \cdot m)$
DP-SGD† (Fernandez-de Retana et al., 2025)	$O(b \cdot m)$	$O(m)$	$O(n \cdot m)$
Adap DP-FL (Fu et al., 2022)	$O(E \cdot b \cdot m)$	$O(m)$	$O(n \cdot m)$
ALDP-FL (Cui and Wu, 2025)	$O(E \cdot b \cdot m)$	$O(m)$	$O(n \cdot m)$
SecAgg (Bonawitz et al., 2017)	$O(n^2 + n \cdot m)$	$O(n + m)$	$O(n^2 + n \cdot m)$
GVSA (Zhou et al., 2025)	$O(g + m)$	$O(g + m)$	$O(n \cdot g + n \cdot m)$
FPMANR (proposed)	$\mathbf{O(E \cdot b \cdot m)}$	$\mathbf{O(m)}$	$\mathbf{O(n \cdot m)}$

7. Conclusion

FPMANR combines FedProx local training, two-neighbor pairwise masking in a ring topology, layer-wise adaptive Gaussian noise, and a client dropout recovery mechanism to protect individual updates and enhance privacy in FL under non-IID data. Experiments on EMNIST Digits, EMNIST Letters, and Fashion-MNIST show that pairwise masking preserves training performance with small computational overhead, while adaptive Gaussian noise gives additional privacy protection with a moderate reduction in accuracy. FPMANR 0.1 achieves test accuracies of 98.10%, 85.39%, and 74.51% on the three datasets and reduces gradient reconstruction success to 0% against DLG, iDLG, and GradInv in all nine dataset–attack combinations, while MIAs remain close to random guessing with AUC values around 0.5. The recovery mechanism keeps aggregation correct under client dropouts without local retraining. Among the evaluated settings, FPMANR 0.1 provides the best balance between privacy and utility, making FPMANR a practical and attack-resistant FL framework for stable training under heterogeneous client data.

CRediT authorship contribution statement

Md. Rashed: Conceptualization, Data curation, Methodology, Visualization, Project administration, and Writing – original draft. **Israt Tasnim Esha:** Data curation, Methodology, Investigation, and Validation. **Dola Das:** Supervision, Formal analysis, Investigation, and Validation. **Nusrat Jahan Jonaki:** Methodology, Formal analysis, Investigation, and Validation. **Ohidujjaman:** Formal analysis, Resources, and Writing – review and editing. **Md. Sarwar Hosain:** Supervision, Formal analysis, Investigation, and Writing – review and editing. **Tetsuya Shimamura:** Supervision, Writing – review and editing, Formal analysis, Investigation, and Funding acquisition.

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Data Availability Statement

The EMNIST and Fashion-MNIST datasets used in this study are publicly available at https://www.nist.gov/itl/products-and-services/emnist-dataset and https://github.com/zalandoresearch/fashion-mnist, respectively. The source code, trained model weights, and experimental results used in this study are available in our GitHub repository at https://github.com/Rashed200613/FPMANR.

Conflicts of Interest

The authors declare that they have no conflicts of interest related to this work.

AI Assistance Statement

AI tools (Claude and Quillbot Premium) were used only to improve grammar, spelling, and language clarity. No part of the research design, analysis, results, or scientific content was generated by these tools. All technical and intellectual contributions in this manuscript were made by the authors.

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